Def.

On an interval I, the differential equation of the form

$$L(y) = a_n \cdot y^{(n)}(t) + a_{n-1} \cdot y^{(n-1)}(t) + \cdots + a_1 y'(t) + a_0 y(t) = b(t)$$

is called the linear equation. If $b(t) \equiv 0$, we say that homogeneous.

Observation.

The L is a differential operator on $C^{(n)}$, because : for any $c \in \mathbb{R}$ and $y_1, y_2 \in C^{(n)}$,

$$L(cy_1 + y_2) = \sum_{i=0}^{n} a_i \cdot (cy_1 + y_2)^{(i)}(t) = \sum_{i=0}^{n} \left[c \cdot a_i y_1^{(i)}(t) + a_i y_2^{(i)}(t) \right] = c \cdot L(y_1) + L(y_2).$$

「Order of 2.

「Thm. Let $a, b \in \mathbb{R}$ be constants, and consider the linear equation

$$L(y) = y'' + a_1 y' + a_2 y = 0.$$

If r_1, r_2 are distinct root of the characteristic polynomial

$$p(r) = r^2 + a_1 r + a_2,$$

then

$$\phi(x) = c_1 e^{r_1 x} + c_2 e^{r_2 x} \quad (c_1, c_2 \in \mathbb{R})$$

is the general solution for $L(y) = 0$.

If r_1 is a repeated root of P, then

$$\phi(x) = c_1 e^{r_1 x} + c_2 x e^{r_2 x} \quad (c_1, c_2 \in \mathbb{R})$$

is the general solution.

Proof. observe that

$$L(e^{rx}) = r^2 \cdot e^{rx} + a_1 r e^{rx} + a_2 e^{rx} = (r^2 + a_1 r + a_2) \cdot e^{rx} = P(r) \cdot e^{rx}$$

and

$$L(xe^{rx}) = (xe^{rx})'' + a_1 (xe^{rx})' + a_2 (xe^{rx})$$

$$= (re^{rx} + re^{rx} + r^2 x e^{rx}) + a_1 \cdot e^{rx} + a_1 r \cdot x e^{rx} + a_2 x e^{rx}$$

$$= (r^2 + a_1 r + a_2) \cdot x e^{rx} + (2r + a_1) e^{rx}$$

$$= (p(r) \cdot x + p'(r)) \cdot e^{rx}$$

$p(r) = p'(r) = 0$ <=> r is multiple root of P, so Proved.

⌈Non-Homogeneous form.

⌈Thm. If φ_1, φ_2 are two solution of

$$y'' + a_1 y' + a_2 y = 0 \quad \text{on an interval } I \text{ containing } x_0.$$

then $$W(\varphi_1, \varphi_2)(x) = e^{-a_1(x-x_0)} \cdot W(\varphi_1, \varphi_2)(x_0).$$

Proof. We have $\begin{cases} \varphi_1'' + a_1\varphi_1' + a_2\varphi_1 = 0 \\ \varphi_2'' + a_1\varphi_2' + a_2\varphi_2 = 0 \end{cases}$, so multiplying $-\varphi_2$ and φ_1 at first and second, respectively,

and adding: $\begin{cases} -\varphi_1''\varphi_2 - a_1\varphi_1'\varphi_2 - a_2\varphi_1\varphi_2 = 0 \\ \varphi_1\varphi_2'' + a_1\varphi_1\varphi_2' + a_2\varphi_1\varphi_2 = 0 \end{cases}$

$$= \varphi_1\varphi_2'' - \varphi_1''\varphi_2 + a_1(\varphi_1\varphi_2' - \varphi_1'\varphi_2) = 0 \quad \cdots (*)$$

Notice that $(\varphi_1\varphi_2' - \varphi_1'\varphi_2)' = \varphi_1\varphi_2'' + \varphi_1'\varphi_2' - \varphi_1'\varphi_2' - \varphi_1''\varphi_2 = \varphi_1\varphi_2'' - \varphi_1''\varphi_2$

Therefore the $(*)$ is: for $W = W(\varphi_1, \varphi_2)$,

$$W' + a_1 W = 0 \Rightarrow W = C \cdot e^{-a_1 x}$$

Now, $C = W(x_0) \cdot e^{a_1 x_0}$, we obtain the result. ☒

Consider that:

$$L(y) = y'' + a_1 y' + a_2 y = b(x)$$

If ψ, ψ_p are solution of $L(y) = b(x)$, then

$L(\psi - \psi_p) = b - b = 0$, so $\psi - \psi_p$ is a solution of $L(y) = 0$.

Therefore, $\psi - \psi_p = c_1\varphi_1 + c_2\varphi_2$ where φ_1, φ_2 are solution of $L(y) = 0$

So, if we a one 'particular' solution ψ_p, then we get the general solution.

「Particular solution.

Let $\psi_p(x) = u_1(x)\cdot\varphi_1(x) + u_2(x)\cdot\varphi_1(x)$. Then,

$b = \psi_p'' + a_1\psi_p' + a_2\psi_p$ 하 시발 계산

$$= u_1''\varphi_1 + 2u_1'\varphi_1' + u_1\varphi_1'' + a_1u_1'\varphi_1 + a_1u_1\varphi_1' + a_2u_1\varphi_1$$
$$+ u_2''\varphi_2 + 2u_2'\varphi_2' + u_2\varphi_2'' + a_1u_2'\varphi_2 + a_1u_2\varphi_2' + a_2u_2\varphi_2$$
$$= \underbrace{u_1\cdot L(\varphi_1) + u_2\cdot L(\varphi_2)}_{=0}$$
$$+ (u_1''\varphi_1 + u_2''\varphi_2) + 2(u_1'\varphi_1' + u_2'\varphi_2') + a_1(u_1'\varphi_1 + u_2'\varphi_2)$$

Now, if $u_1'\varphi_1 + u_2'\varphi_2 = 0$, then $u_1''\varphi_1 + u_2''\varphi_2 + u_1'\varphi_1' + u_2'\varphi_2' = 0$,

so $u_1'\varphi_1' + u_2'\varphi_2' = b$. is remained

Then, $u_1' = \dfrac{-\varphi_2 b}{W(\varphi_1,\varphi_2)}$ and $u_2' = \dfrac{\varphi_1 b}{W(\varphi_1,\varphi_2)}$, so

$$\psi_p(x) = \int_{x_0}^{x} b(t)\cdot\begin{vmatrix}\varphi_1(t) & \varphi_1(x)\\ \varphi_2(t) & \varphi_2(x)\end{vmatrix}\cdot\frac{1}{W(\varphi_1,\varphi_2)(t)}\,dt.$$